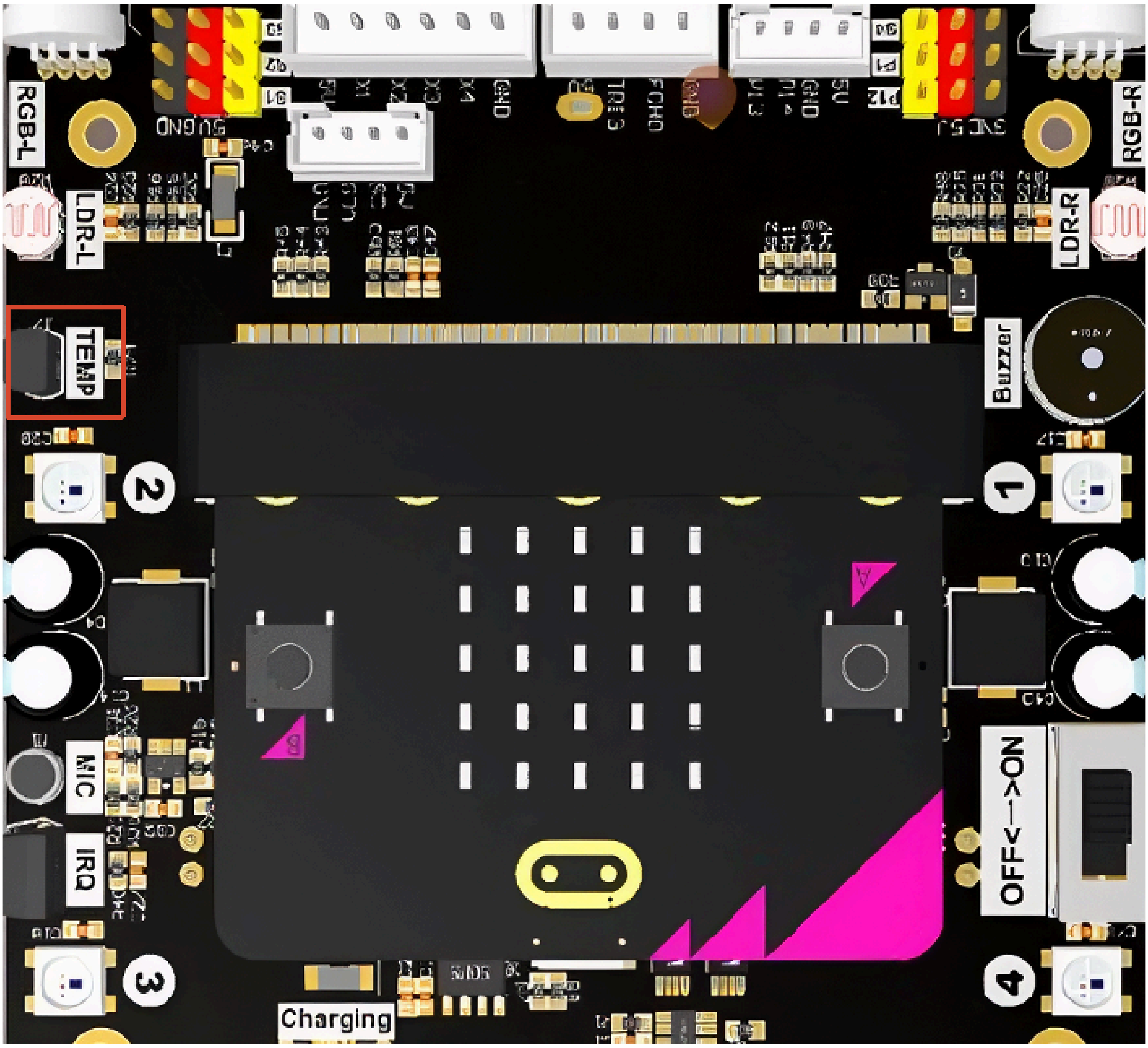
Temperature Collection

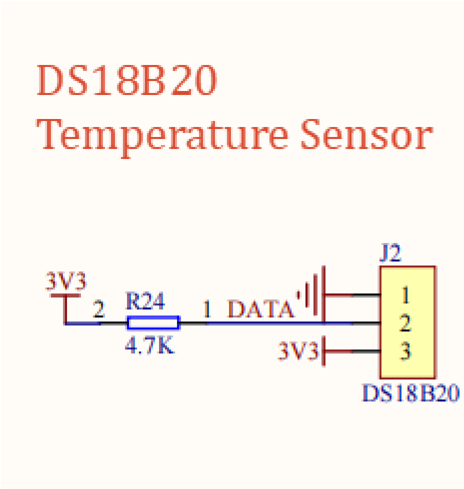
1. Experiment Purpose

The Lastbit car is equipped with a DS18B20 temperature sensor on board, with the silk screen marking `TEMP`. This lesson will guide you through learning how to read temperature data and display it on the Micro:bit LED matrix.



The DS18B20 is a digital temperature sensor produced by Dallas Semiconductor (Maxim). Its appearance resembles a transistor. Its main features are as follows:

- **Digital Signal Output**: Directly outputs digital temperature values without ADC conversion
- **1-Wire Single-Bus Communication**: Only one data line is needed to communicate with the main controller
- **Measurement Range**: -40°C to +125°C
- **Measurement Accuracy**: ±0.5°C (within the range of -10°C to +85°C)
- **Programmable Resolution**: 9 to 12 bits configurable
- **Unique Serial Number**: Each chip has a unique 64-bit serial number, supporting multiple sensors mounted on the same bus
- **Multi-Point Networking**: Multiple DS18B20s can be connected in parallel on the same data line for separate readings
- **Parasitic Power Supply**: Supports drawing power from the data line

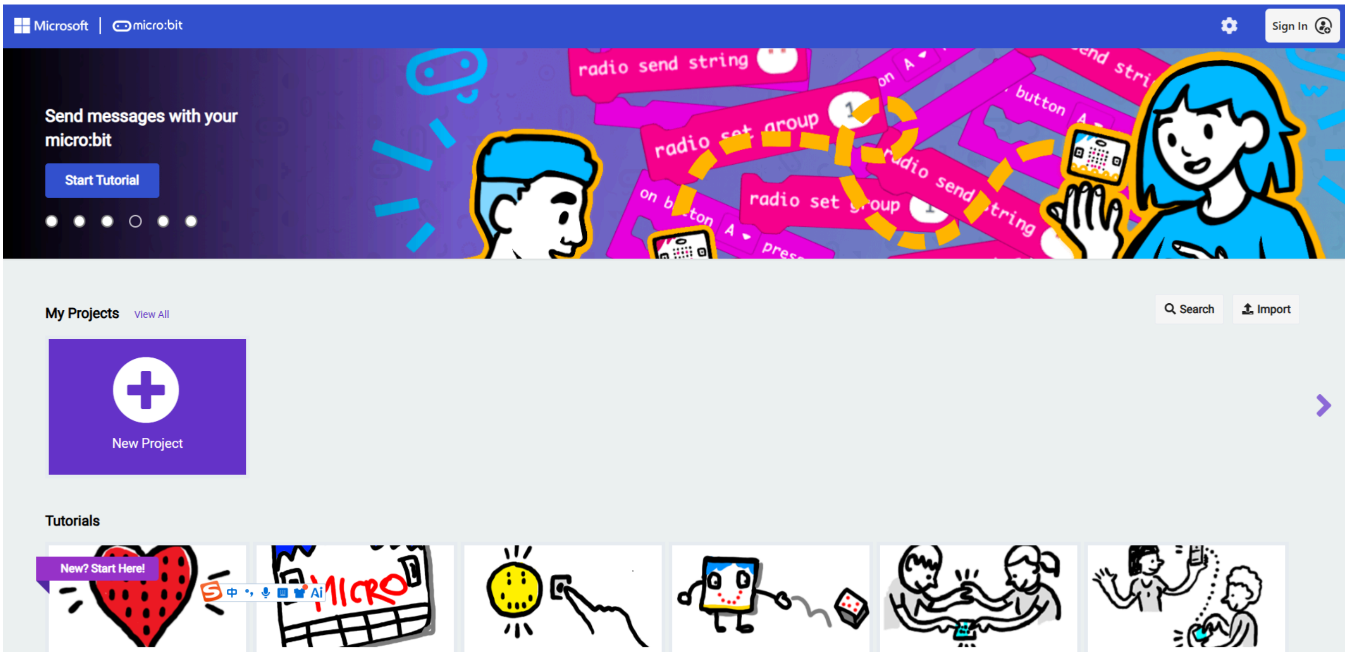


2. Experiment Steps

1. Before starting the experiment, connect the Micro:bit mainboard to your computer using a Micro USB data cable. At this time, the computer will display the MICROBIT drive letter.

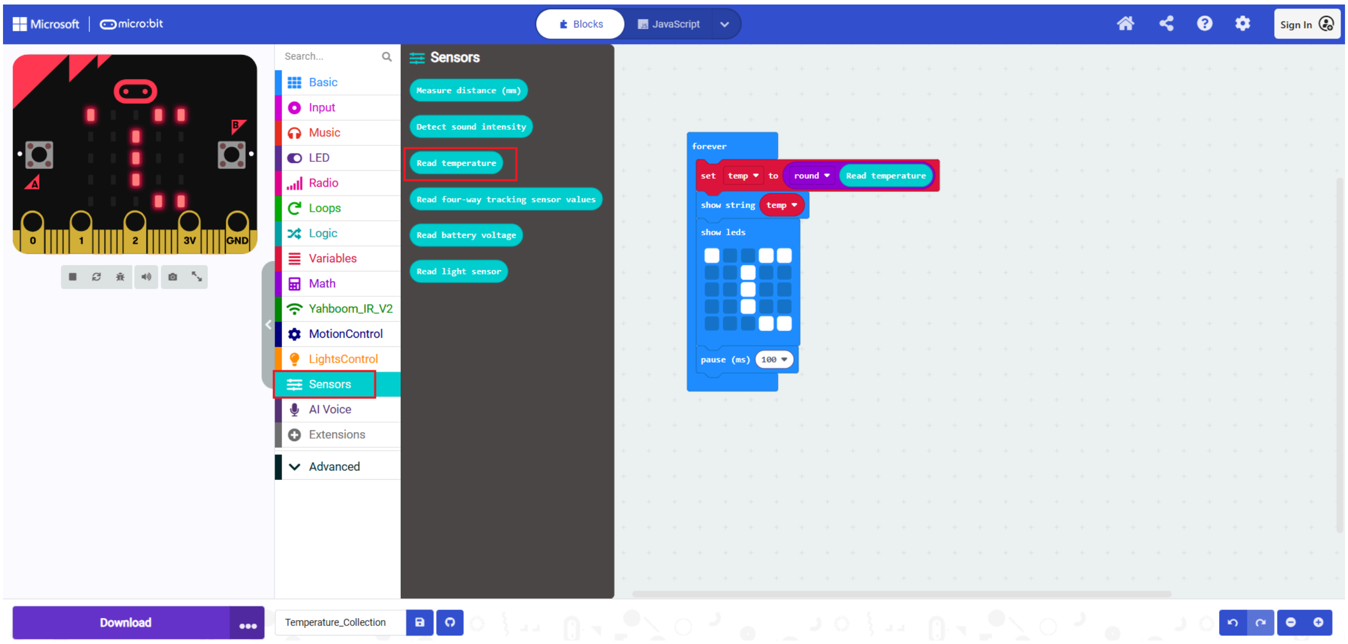
2. Open MakeCode using the link below.

   [Microsoft MakeCode for micro:bit]



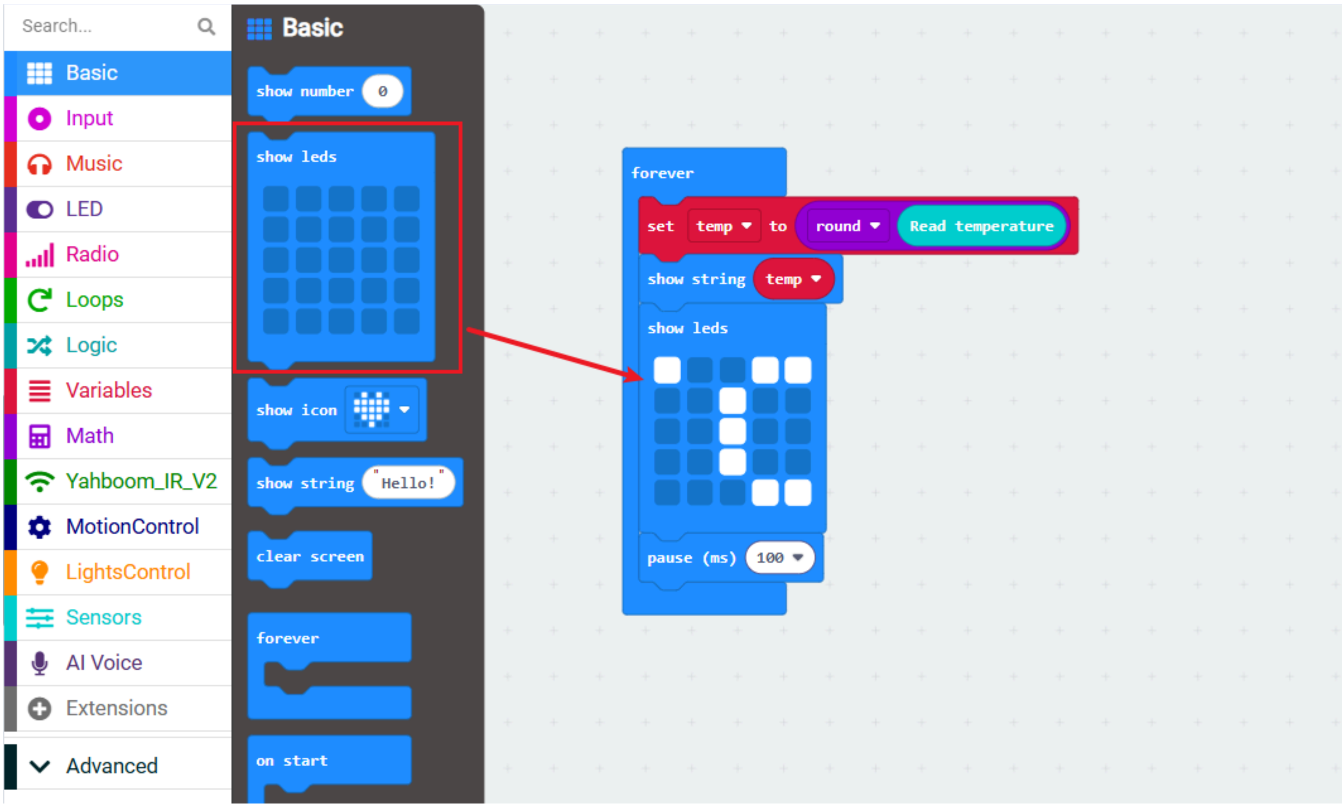
3. Drag `5.Temperature_Collection.hex` to the new project page. MakeCode will automatically import and open the project.

3. Experiment Source Code

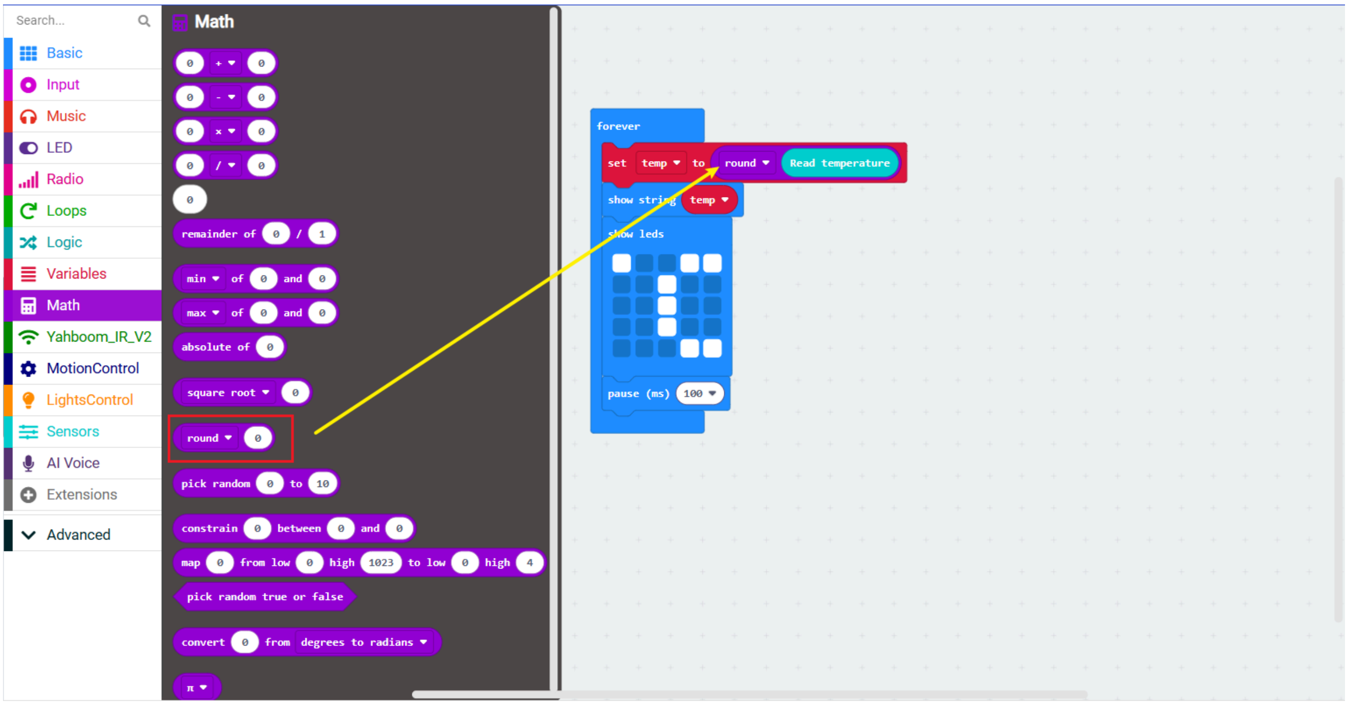


The building blocks used in this lesson are located in the `Sensors` category.

The building block for displaying the (℃) symbol is located in the `Basic` category. Drag the `Show leds` to the code editing area to draw the pattern you want to display on top.

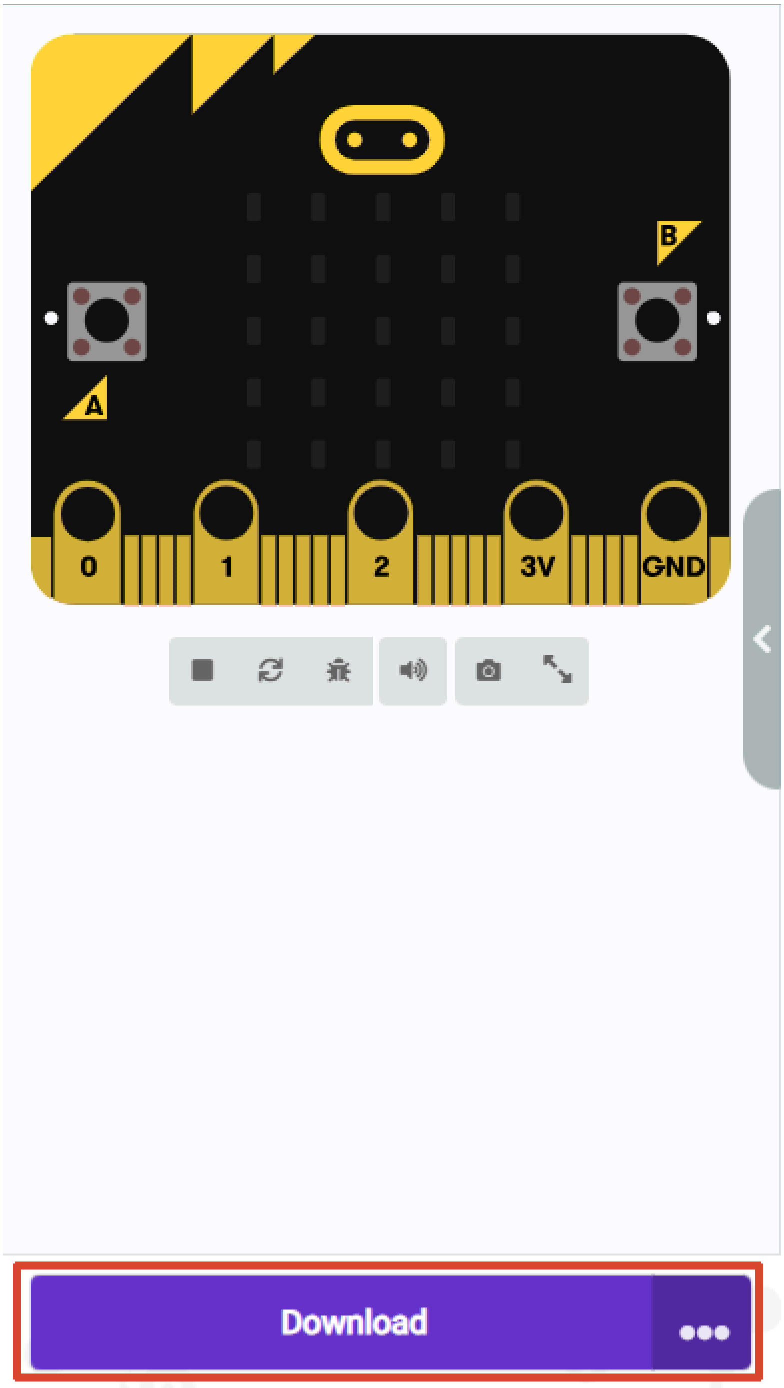


The rounding block is located in the `Math` category. Take the integer value to display on the LED matrix.

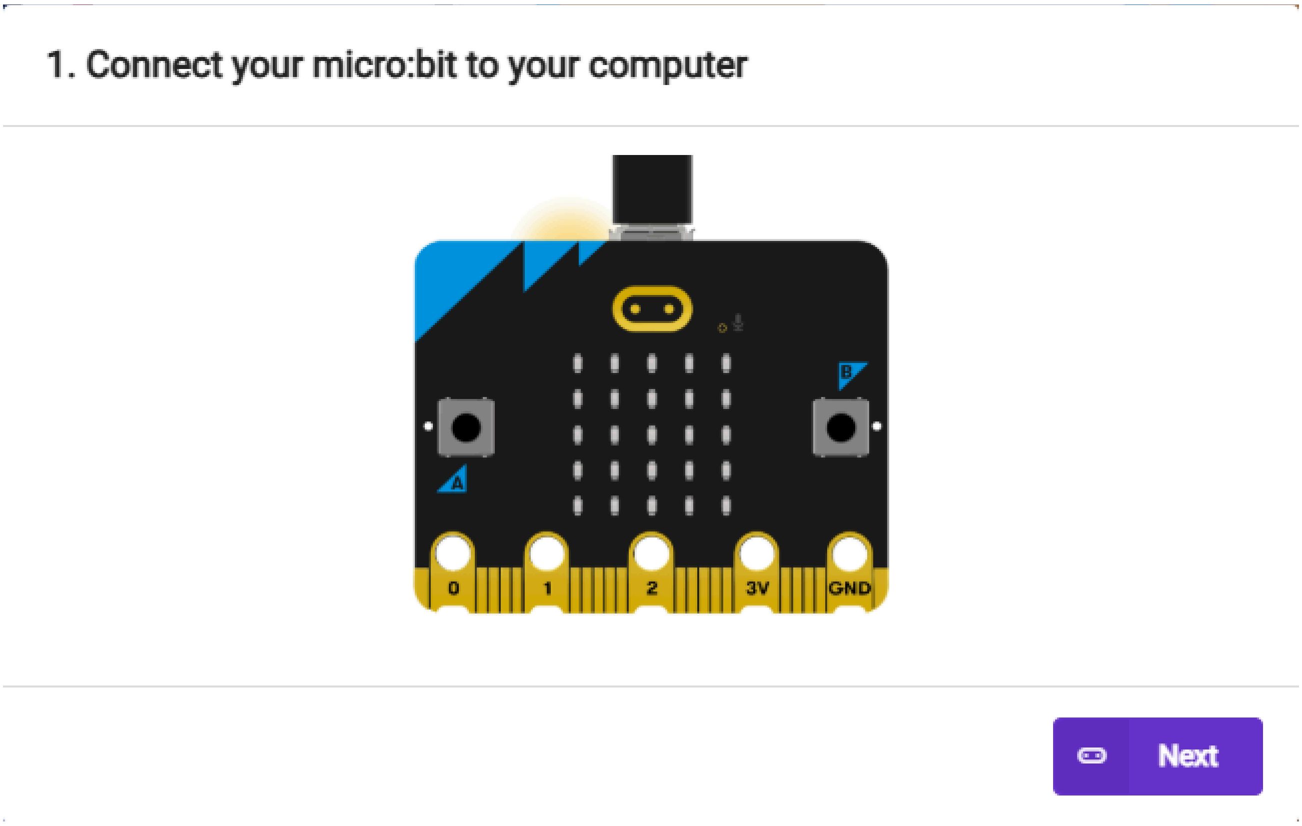


4. Program Download

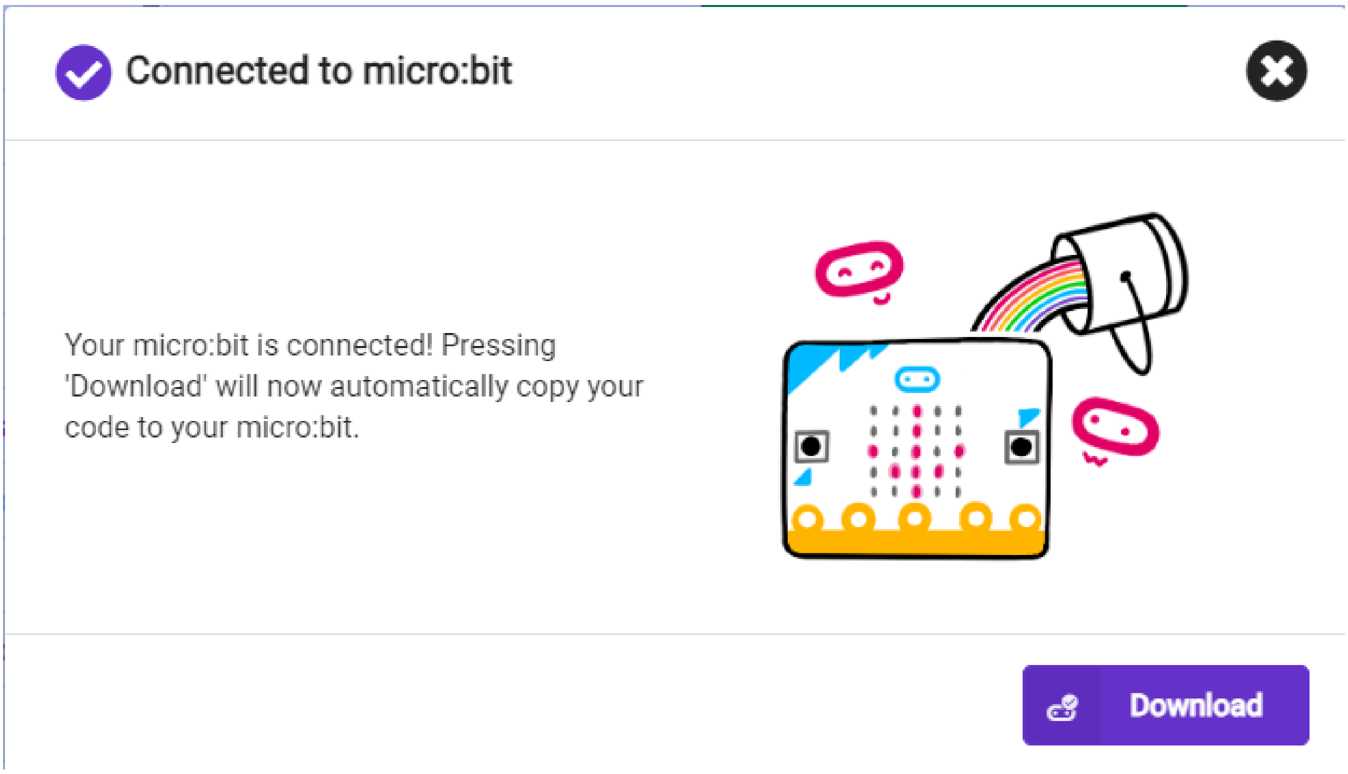
1. Click the download button at the bottom left to download the program to the Micro:bit.



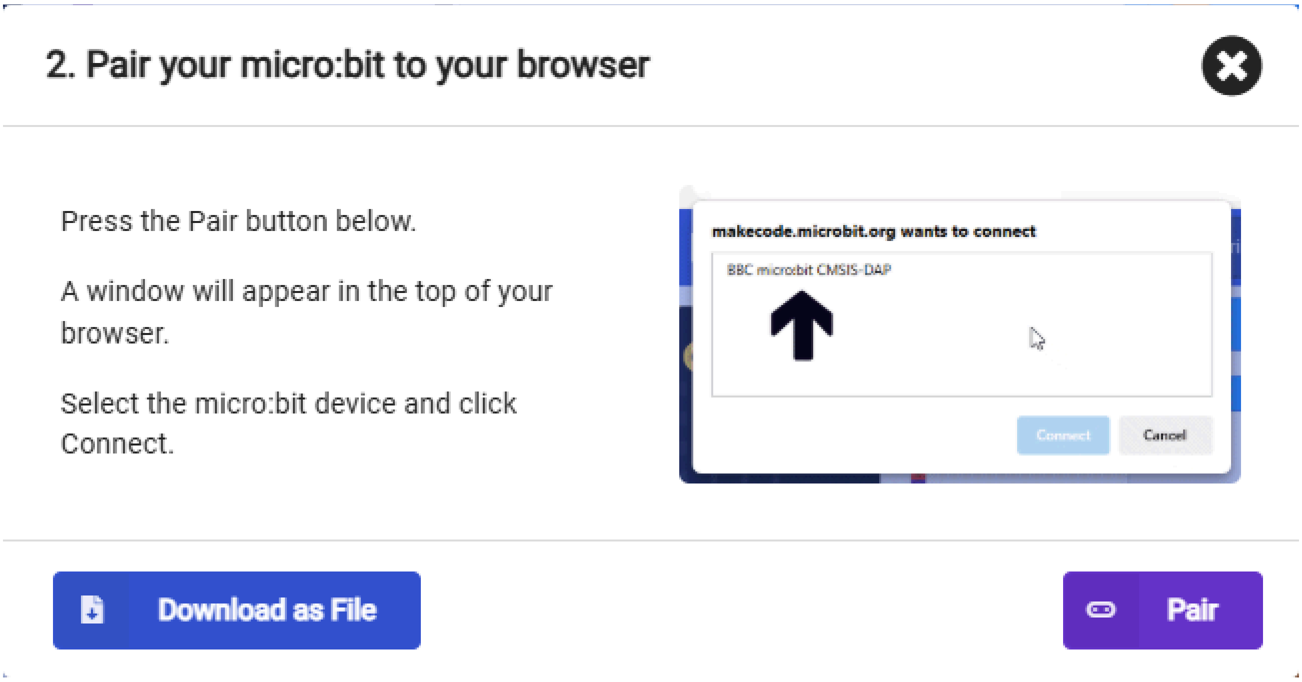
2. When the Micro USB data cable is not connected to the Microbit, the system will first prompt you to connect the device. If not connected, please connect the data cable to the Microbit first. If already connected, click [Next] here.



3. If the Microbit is detected as properly connected, the following interface will be displayed. At this point, you can click [Download].

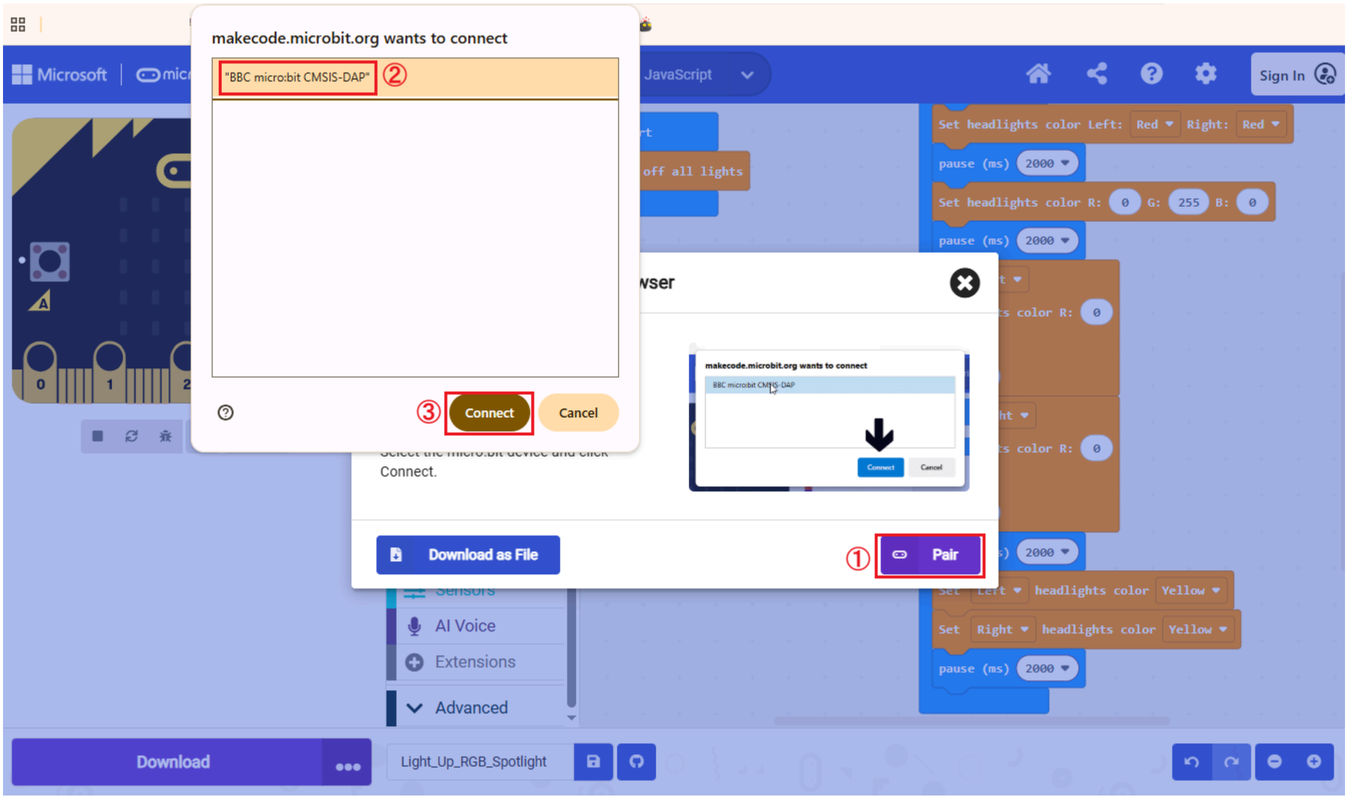


If the Microbit data cable is not properly connected or multiple Microbits are connected, the system will require you to manually select the device. You can choose between two methods: [Download as File] and [Pair].

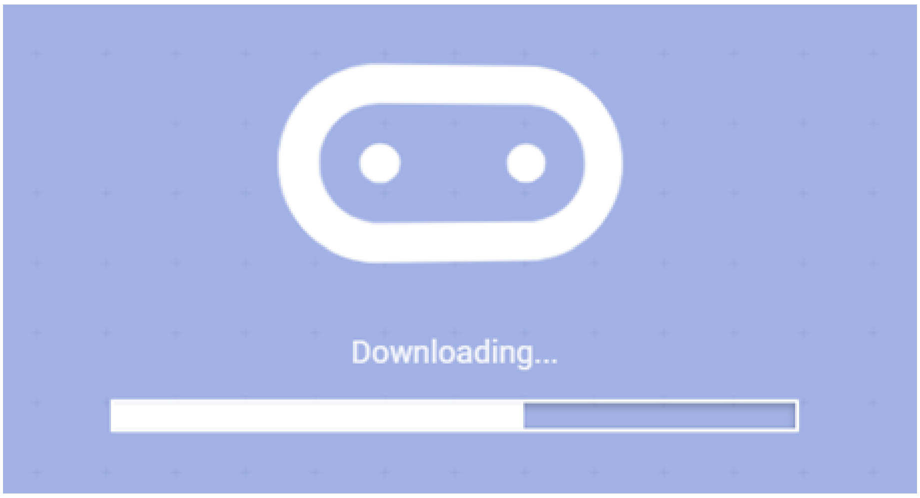


When selecting [Download as File], the system will download the currently opened program as `microbit-project-name.hex`. You can directly copy or drag this program to the corresponding MICROBIT drive letter to complete the download.

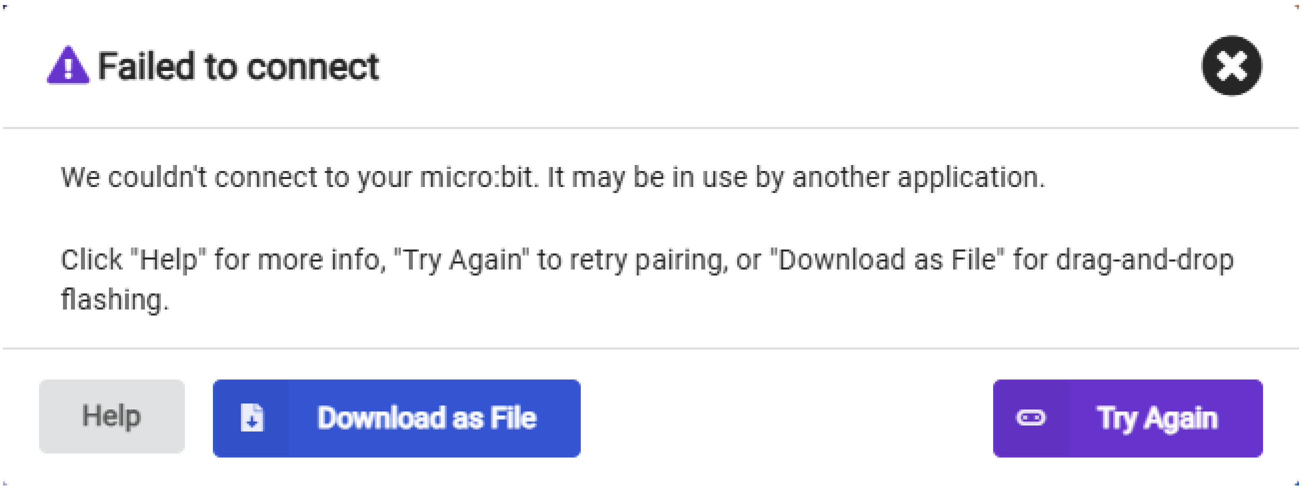
When selecting [Pair], you will be asked to manually select the device. Here, select `BBC mico:bit CMSIS:DAP`.



4. When the following interface appears, it means the program is being downloaded. Wait for the progress bar to complete to download the program to the Microbit.



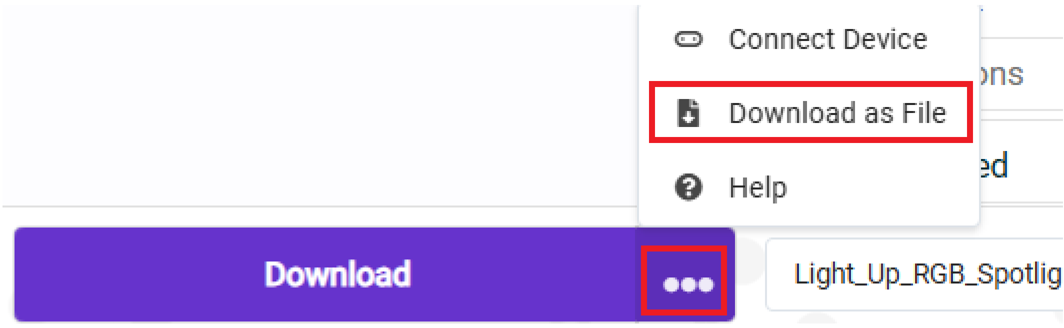
If multiple attempts continue to fail, please directly select [Download as File], save the current program as a file, and directly copy or drag the program to the corresponding MICROBIT drive letter to complete the download.



If you want to save the file directly, we provide two methods:

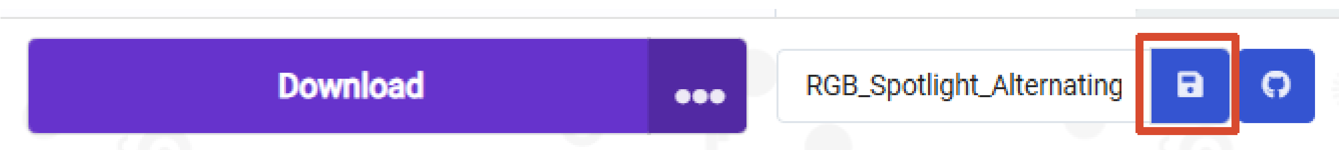
Method 1:

Click the [...] next to download, then select [Download as File]



Method 2:

Click [Save] after the [Project Name Edit Field].



5. Experiment Results

After flashing, turn on the switch (flip the switch to the ON position). Hold the temperature sensor's black probe with your hand (note: do not touch the mainboard with wet hands to avoid short circuits). The Micro:bit screen will display the read temperature value. The temperature value will slowly rise. After releasing your hand, the temperature will slowly drop.